

some market participants with a preview of whether there would be a match before the entire marketplace. Many believed that this provided an unfair advantage to those privileged to these flash orders. Flash trading is no longer allowed in any of the market venues.

TYPES OF ORDERS

The market allows numerous different types of orders such as market, limit, stop loss, etc. But the three most important order types for algorithmic trading are market, limit, and marketable limit orders.

Market Order: A market order specifies to the algorithm to buy or sell at the best available market price. This order is most likely to be executed because there are no restrictions on its price and it will not be placed into an order book. The disadvantage is that in today's markets, prices can move away so quickly that the best ask or best bid could in effect be much higher or much lower than they were at the time the order was routed for execution. Market order will "take" liquidity.

Limit Order: A limit order specifies to the algorithm to buy or sell at the specified limit price or better. In most cases the limit order will be entered into the order book of the exchange or venue and is subject to the queue before it is eligible to be executed. For example, in price-time priority, existing orders at that price or better will need to transact before that order with an offsetting buyer. A limit order is not guaranteed to execute, but provides some safety surrounding the execution price and ensures that the execution will not be worse than the pre-specified limit price. A limit order will "provide" liquidity and is also referred to as posting an order.

Marketable Limit Order: A marketable limit order is an order that specifies to the algorithm to buy or sell at a specified price or better. This order will either be executed in the market at the specified price or better, or be cancelled if there are no existing orders at that price or better in the market.

Rebates: Depending upon the exchange or venue, investors may receive a rebate for posting liquidity to the exchange and others may provide a rebate for taking liquidity from the exchange. Different rebate models are discussed further in Chapter 2.

EXECUTION OPTIONS

Here the investor provides the broker with the order or basket to trade in the market on their behalf. The broker exerts "best efforts" to achieve the best prices for the investor. They receive a commission for their role in